

Research Methods: Quantitative & Qualitative Design

Part 1 of the Research Methods Professional Certificate · 10 units · 20 contact hours · Beginner to Intermediate

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A complete foundation in research methods, from framing an answerable question through reviewing the literature, choosing a design, sampling, building instruments and meeting ethical obligations, to writing a proposal that will be approved. Quantitative and qualitative approaches are taught side by side rather than as separate worlds, because most real research questions require judgement about which one — or both — actually fits.

Who it is for

- Postgraduate students preparing a thesis or dissertation proposal
 - Early-career researchers, research assistants and research nurses
 - Health, social science and laboratory professionals starting a study
 - Programme and M&E staff who commission or interpret research
 - Anyone who has been asked to "just add a methods section"
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Prerequisites

- No prior statistics, software or research experience is required
 - Bring a research idea if you have one — the activities build towards a proposal
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Learning outcomes

- State a research question with clear aims, objectives and scope
- Conduct and document a reproducible literature search across the major databases
- Manage references and generate a bibliography automatically
- Choose a quantitative, qualitative or mixed design and justify it against alternatives
- Select a sampling strategy and calculate or defend your sample size
- Build or adapt an instrument and evidence its validity and reliability
- Meet the ethical obligations of research with human participants
- Write a proposal that an ethics committee and a supervisor will accept

Schedule of units

TEN UNITS

#	Unit	Duration	Focus
1	Foundations of Research	90 minutes	What separates research from other ways of knowing, the main types of research, and the philosophical positions that quietly shape every methodological choice you will make later.
2	The Research Problem, Questions and Objectives	90 minutes	How to move from a broad interest to a question that is specific enough to answer, and how aims, objectives, hypotheses and conceptual frameworks fit together.
3	The Literature Review	120 minutes	What a literature review is for, the main types and when each is appropriate, and how to build one that argues rather than merely lists.
4	Literature Databases, Searching and Reference Management	150 minutes	The databases that matter, how to build a search that is both sensitive and precise, how to document it for PRISMA, and how to manage references so the bibliography is never done by hand.
5	Quantitative Research Designs	120 minutes	The main observational and experimental designs, what each can and cannot establish, and how to choose the weakest design that still answers your question honestly.
6	Qualitative Research Designs	120 minutes	The major qualitative traditions, the data collection methods attached to each, and how rigour is established when the aim is understanding rather than measurement.
7	Mixed Methods and Sampling	120 minutes	How quantitative and qualitative strands are combined deliberately rather than decoratively, and the full range of sampling strategies with the arithmetic of sample size.
8	Data Collection, Instruments and Measurement	120 minutes	Turning abstract constructs into things you can actually record, and the validity and reliability evidence needed before anyone believes the numbers your instrument produces.
9	Research Ethics and Integrity	120 minutes	The principles governing research with human participants, how ethical review actually works, and the integrity questions — authorship, plagiarism, data fabrication and the use of AI — that decide whether your work stands.
10	Writing the Proposal, Rigour and Dissemination	120 minutes	Assembling everything into a proposal that will be approved and funded, building in rigour rather than claiming it, and planning how findings will reach the people who can act on them.

Assessment

Assessment is continuous and formative, ending in a short proposal that you can take directly into your own work.

COMPONENTS

Component	Weight	Detail
Unit quizzes	30%	A self-check quiz after each of the ten units, with immediate feedback and an explanation for every option. Unlimited attempts.
Unit activities	30%	A short practical task per unit, built around your own research idea and reviewed in pairs during the live session.
Final proposal	40%	A two-page proposal skeleton: title, aim, objectives, design, sampling and sample size, analysis plan and limitations. Marked against the rubric below with written feedback.

FINAL PROPOSAL MARKING RUBRIC

Criterion	Weight	What earns the marks
Problem statement and gap	15%	Magnitude evidenced; gap specific and of an identifiable type.
Question, aim and objectives	20%	Answerable question; objectives measurable and mapping to the analysis.
Design and justification	20%	Design fits the question; alternatives considered and rejected with reasons.
Sampling and sample size	20%	Strategy appropriate; calculation shown with all inputs, or saturation logic defended.
Ethics and rigour	15%	Consent, confidentiality and risk addressed; limitations state direction and mitigation.
Clarity and structure	10%	Coherent, concise, correctly referenced.

Pass mark: 60% overall.

Certificate

ICBB Certificate of Completion — Research Methods (Part 1). Awarded on achieving 60% overall and submitting the final proposal.

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